

SEDAR 680A Atlantic Scamp

- Results of this assessment are sensitive to natural mortality. The scale and age dependence of natural mortality were estimated using meta-analytical methods, as is common in SEDAR assessments. While such methods describe relationships between M and other life-history characteristics (growth, maximum age) averaged across species, they may not describe well the natural mortality of any particular species. Mark-recapture approaches (conventional, telemetry, close-kin) might make it possible to obtain direct estimates of natural mortality specific to scamp in the South Atlantic region.
- More research on the usage and effect of descender devices would benefit stock assessments of South Atlantic reef fishes.
- More research is needed on the cause(s) of low recruitment in several South Atlantic reef-fish stocks, including scamp. This topic is currently being investigated by the SEFSC.
- Better characterize reproductive parameters including age at maturity, sex transition, batch fecundity, spawning seasonality, and spawning frequency. Mature male and female biomass was the measure of reproductive potential in the assessment, to account for the importance of males in a protogynous hermaphrodite. However, this approach models the potential for sperm limitation implicitly, and more direct modeling approaches could be developed. Theoretical development would be desirable, but we note that for use in assessments, new modeling methods might also require the support of new data collections (e.g., to quantify sperm limitation as a function of sex ratio.)